

Hitec University Taxila, Department of Computer Engineering

Lab Manual for Mobile Application Development

Lab-5 List View and Array Adapter

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1 Introduction

Android ListView is a view which groups several items and display them in vertical scrollable list. The list items are automatically inserted to the list using an Adapter that pulls content from a source such as an array or database. Android provides several subclasses of Adapter that are useful for retrieving different kinds of data and building views for an AdapterView (i.e. ListView or GridView) The common adapters are

- ArrayAdapter
- BaseAdapter

You can use this adapter when your data source is an array

By default, ArrayAdapter creates a view for each array item by calling toString() on each item and placing the contents in a TextView. Consider you have an array of strings you want to display in a ListView, initialize a new ArrayAdapter using a constructor to specify the layout for each string and the string array

```
ArrayAdapter adapter = new ArrayAdapter(this, R.layout.history_item_layout,  
countryArray);
```

2 Activity Time boxing

Table 1: Activity Time Boxing

Task#	Activity Name	Activity time	Total Time
1	Introduction	60mins	60min
2	Walk Through Task	60mins	60min
3	Evaluation Task	60min	60min

3 Objectives of the experiment

- To get basic understanding List View and Adapter
- To be able to use List View and Adapter in android applications.
- To get to know about the benefits related to List View.

4 Walk through Task

4.1.1 Creating New Activity

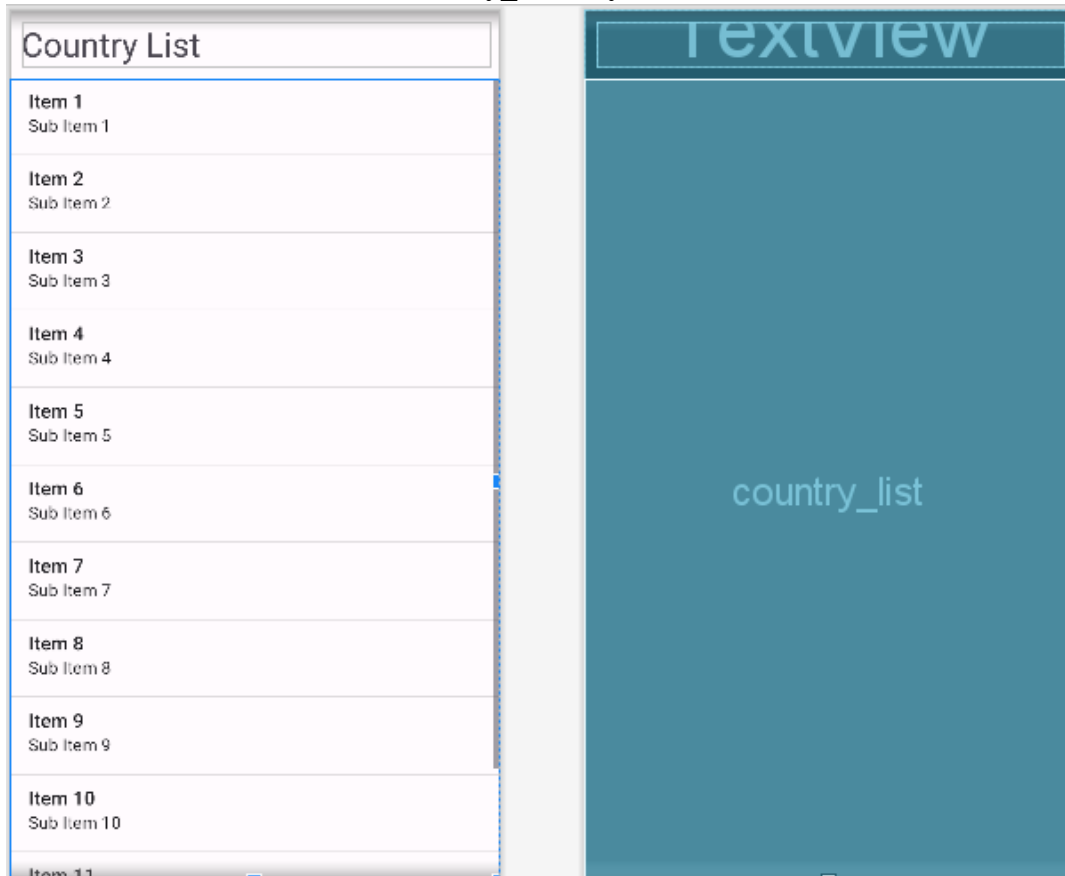
Step 1: Create Activities:

Create a new project just like you have created in your previous labs and create one more empty activity.

Step 2: Design XMLs of Activity:

Designs of activity are given below, just use the given xml code which is given in this step.

activity_country.xml



```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical">

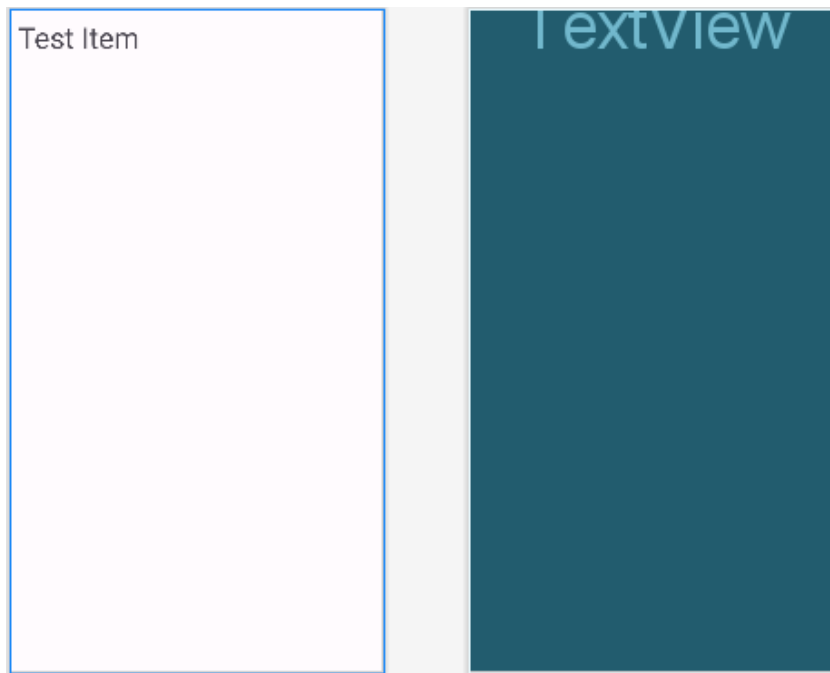
    <TextView

        android:id="@+id/list_tv"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_margin="@dimen/view_margin"
        android:textSize="@dimen/font_size"
        android:text="Country List" />

    <ListView
        android:id="@+id/country_list"
        android:layout_height="match_parent"
        android:layout_width="match_parent">
    </ListView>

</LinearLayout>
```

Step 3: Design country_item_view.xml:



```
<?xml version="1.0" encoding="utf-8"?>
<TextView
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:padding="@dimen/view_margin"
    android:text="Test Item"
    android:textSize="32sp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">
</TextView>
```

Step 4: Write adapter in Main Activity

```
public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_country);

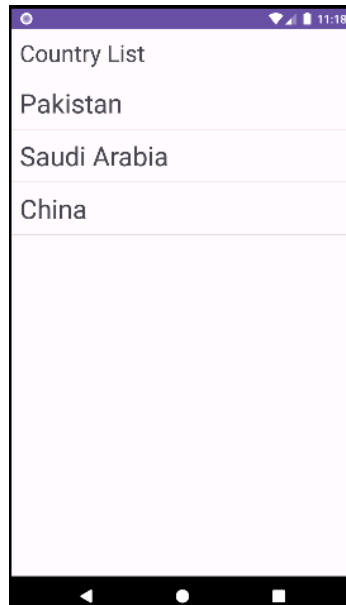
        ListView countryList = findViewById(R.id.country_list);

        ArrayList<String> countryArray = new ArrayList<String>();
        countryArray.add("Pakistan");
        countryArray.add("Saudi Arabia");
        countryArray.add("China");

        ArrayAdapter adapter = new ArrayAdapter(this,
        R.layout.country_item_layout, countryArray);
        countryList.setAdapter(adapter);
    }
}
```

4.1.2 Step 6: Now Run Project:

Click “Run” or press “Shift + F10”.



5 Home Task

Consider your assignment # 1 and implement the same task using list view on history activity and use array list to add count in main activity and pass this array list to history activity.

Hint:

```
ArrayList<String> historyAL =  
getIntent().getStringArrayListExtra(TasbeeActivity.MAIN_TO_HISTORY_STRING_KEY);
```

